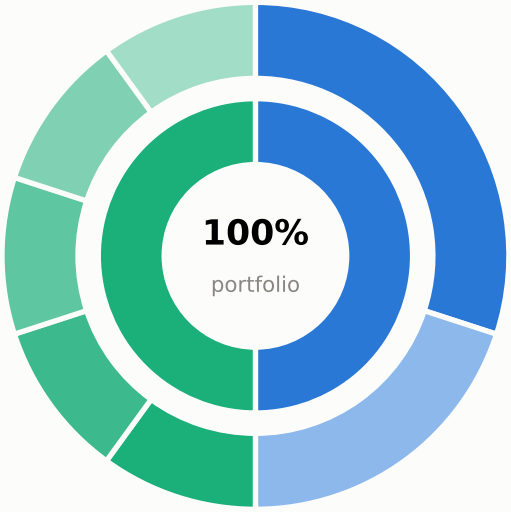
STRATEGY HANDBOOK

Core & Shield

How it works, why it exists, and how to judge it

Version 1.0 · 27 September 2026 · written before the first trade (1 October 2026)

> **In one sentence:** half the money sits in a classic 60/40 (the **Core**), the other half is run by a rules-based model (the **Shield**) that each month picks the strongest, calmest and least-crowded of 12 ETFs, or retreats to cash. The goal is roughly a 60/40's return with much smaller losses when markets fall.



- **Core · 50%** · fixed, never changes
  - SPY 30% · US stocks (S&P 500)
  - AGG 20% · US bonds
- **Shield · 50%** · RAAM, re-picked monthly
  - 5 slots × 10% each, chosen from 12 ETFs
  - a slot goes to cash (BIL) if its ETF is weaker than cash
  - Bitcoin (IBIT) never more than 5% of the whole portfolio

Backtest, Feb 2015 – Sep 2026, after trading costs (Bitcoin before 2024 simulated; see section 6)

Return per year	Sharpe ratio	Worst month-end drop	2022 (bear market)
9.3%	**1.18**	**−11.1%**	**−8.8%**
60/40: 9.2%	60/40: 0.93	60/40: −19.2%	60/40: −15.2%

At a glance	
Account	Alpaca paper trading account, starting value $100,000 (move to IBKR paper planned)
Trades	Once a month: signal after the last trading day's close, orders the next morning
Run by	Claude Code cloud routines; the maths and orders are a fixed Python script
Alerts	Discord: daily health line, monthly PDF report, any halt
Status	Locked for evaluation from 24 Sep 2026; judged against 60/40 at 1, 3, 6 and 12 months

1 The idea

Why this strategy exists, where it comes from, and what it is (and isn't) for.

The problem it solves

A 60/40 portfolio is the standard balanced benchmark, but its losses in a bad year are large: **−15% in 2022** and about **−19%** at its worst month-end in our test window. Tactical rotation models are the opposite: they sidestep much of a slow bear market by moving into bonds or cash, but they lag badly in strong bull markets because they hold defensive assets too often.

Core & Shield combines the two. The **Core** is a fixed 60/40 that captures most of the market's long-run growth. The **Shield** is a rotation model that behaves like a shock absorber. Together they aim for **about the same return as a 60/40, with roughly half the damage in bear markets**.

Where it comes from

The Shield is the **Ranked Asset Allocation Model (RAAM)** by Gioele Giordano, which won the 2018 Charles H. Dow Award from the CMT Association. RAAM ranks a diversified set of ETFs on momentum, volatility and correlation each month and holds the best five, moving to cash when nothing is trending up.

Our version is **not** the paper's version. The paper's headline (+638%, Sharpe 1.94 over 2004–17) was in-sample, before costs, and could not be reproduced. We resolved the paper's seven ambiguities, made three safety upgrades, removed a factor that hurt results, wrapped it in the Core, and added a small, capped Bitcoin sleeve. Section 9 lists every change and the evidence behind it.

What it is designed to do

- **Specialise in defence.** It wins mainly by losing less, not by making more. Its best years relative to 60/40 are bad years for markets (2018 aside, see section 6).
- **Stay invested for growth.** Half the portfolio is always in stocks and bonds, so it does not miss recoveries the way a pure timing model can.
- **Follow rules, not opinions.** The allocation is computed by a script. Nobody, including Claude, overrides it because of news or a market view.
- **Trade rarely.** One rebalance a month, about 15% of the portfolio turned over each time.

What it is not

- **Not a way to beat the S&P 500.** In a strong bull market it will trail SPY, often by a lot, and usually trail 60/40 a little. That is by design.
- **Not a crash-proof product.** A sudden one-day crash hits before any monthly model can react; the circuit breaker limits the damage but cannot prevent it.
- **Not a Bitcoin strategy.** Bitcoin is one candidate of twelve and is capped at 5%.

Who a strategy like this suits

Investors who care more about the size of the worst loss than about maximum return: people drawing an income, those near a goal date, and institutions such as pension and superannuation funds that are judged on risk-adjusted returns. For them, matching a 60/40's return with a much smaller drawdown is a meaningful improvement.

> **Its personality in one line:** a balanced portfolio that expects to be roughly level with 60/40 in good years and clearly ahead in bad ones.

2 How the portfolio is built

Two halves, 13 possible holdings, one cap.

Half	Weight	Holdings	How it changes
Core	50%	SPY 30% + AGG 20%	Never. Reset to exactly these weights at every monthly rebalance.
Shield	50%	5 slots of 10% each, from 12 candidate ETFs, with BIL (cash) for failed slots	Re-chosen every month by the RAAM rules in section 3.

The two halves simply add up. If the Shield picks AGG, the portfolio holds 20% (Core) + 10% (Shield) = **30% AGG**. If the Shield sends a slot to cash, that 10% sits in BIL, a 1–3 month Treasury bill fund that earns the short-term interest rate.

The Shield's 12 candidates

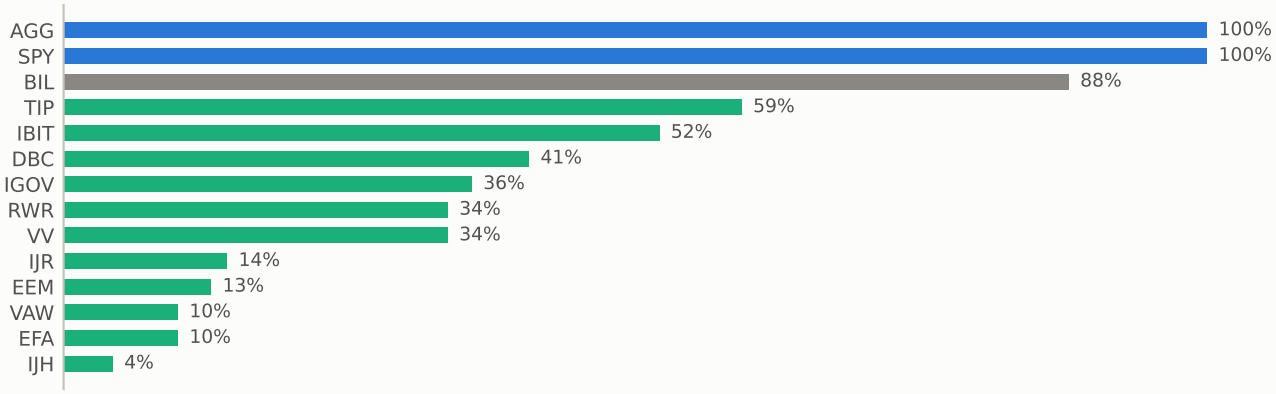
Eleven ETFs from Craig Israelsen's **7Twelve** diversified portfolio, plus spot Bitcoin. Together they span stocks, real estate, commodities, bonds and crypto, so there is usually something trending up.

Asset class	ETF	What it holds
US stocks	VV	Vanguard Large-Cap: large US companies
	IJH	iShares Core S&P Mid-Cap: mid-sized US companies
	IJR	iShares Core S&P Small-Cap: small US companies
International stocks	EFA	iShares MSCI EAFE: developed markets outside North America
	EEM	iShares MSCI Emerging Markets
Real estate	RWR	SPDR Dow Jones REIT: US real estate investment trusts
Commodities / resources	DBC	Invesco DB Commodity Index: energy, metals, agriculture
	VAW	Vanguard Materials: mining, chemicals and materials companies
US bonds	AGG	iShares Core US Aggregate Bond: the broad US bond market
	TIP	iShares TIPS: inflation-protected US Treasuries
International bonds	IGOV	iShares International Treasury: developed-market government bonds
Crypto	IBIT	iShares Bitcoin Trust: spot Bitcoin. **Capped at 5% of the portfolio**
Cash (not ranked)	BIL	SPDR 1–3 Month T-Bill: where failed slots and any IBIT excess go
Core only	SPY	SPDR S&P 500: the Core's stock holding (not a Shield candidate)

The Bitcoin cap

A normal Shield slot is 10% of the portfolio. If IBIT is picked, it only gets **5%**; the other 5% goes to BIL. The cap is enforced by the script after the ranking, so the portfolio never starts a month with more than 5% in Bitcoin (between rebalances it drifts with Bitcoin's price). A Bitcoin crash of −80% while it is held costs roughly 3–4% of the portfolio.

How often each ETF was held (backtest, % of months)



Blue = always held through the Core. Grey = cash. SPY and AGG show 100% because the Core always holds them; the Shield also picks AGG in some months.

3 The monthly decision

Eight mechanical steps, computed by the script after the last trading day's close.

1 Measure
M, V, C for all 12 ETFs at month-end

2 Rank
1 (best) to 12 on each factor

3 Score
average of the three ranks

4 Pick 5
the five lowest scores, 10% each

5 Cash test
a pick weaker than cash goes to BIL

6 Add the Core
+ SPY 30%
+ AGG 20%

7 Cap Bitcoin
IBIT ≤ 5%, excess to BIL

8 Trade
next morning, then hold all month

The three factors

Factor	Measured as	Best rank goes to	Why it matters
M · Momentum	Price change over the last 84 trading days (about 4 months)	Highest momentum	Assets that have been rising tend to keep rising for a while
V · Volatility	Daily price swings (EWMA, λ = 0.94, annualised, smoothed over 10 days)	Lowest volatility	Calmer assets give smoother returns and smaller losses
C · Correlation	Average correlation of daily returns with the other 11 candidates over 84 days	Lowest correlation	Assets that move differently add real diversification

Score and selection

Score = ⅓ × Rank(M) + ⅓ × Rank(V) + ⅓ × Rank(C) − Momentum ÷ 1000 (lower is better)

- Each candidate is ranked 1 (best) to 12 on each factor; ties share the average rank.
- The tiny momentum term only breaks exact ties, in favour of the stronger trend.
- The **five lowest scores** are picked. Each gets 20% of the Shield, which is 10% of the portfolio.
- **Cash test:** a pick keeps its slot only if its 84-day momentum beats cash's own (BIL's) 84-day momentum. Otherwise that slot goes to BIL. If nothing beats cash, the whole Shield (50%) sits in BIL while the Core stays invested.
- A fourth factor from the paper (an ATR trend/breakout signal) is still calculated and logged, but **no longer scored**: three separate backtests showed results were better without it.

Worked example (illustrative numbers)

Suppose the five lowest scores are TIP, IBIT, DBC, AGG and VV, cash momentum is +1.3%, and VV's momentum is −1.0%:

Step	SPY	AGG	TIP	DBC	IBIT	VV	BIL	Total
Shield picks (10% each)		10%	10%	10%	10%	10%		50%
Cash test: VV −1.0% < cash +1.3%		10%	10%	10%	10%	0%	10%	50%
Add the Core	30%	30%	10%	10%	10%		10%	100%
Cap IBIT at 5%	30%	30%	10%	10%	**5%**		**15%**	100%

The last row is what gets bought on the first trading day and held for the month.

4 Timing and operations

Who does what, and when. Everything runs in the cloud; the GitHub repository is the system's memory.

A month in the life

When (US Eastern)	What happens
Every weekday after the close (17:30 summer / 16:30 winter)	Health check: account value, drawdown, positions vs targets, circuit breaker. One Discord line.
Last trading day, after the close	Research scan first (context only), then the **signal** is computed and saved. Pre-flight check that every ETF is tradable.
Last trading day, 19:00 summer / 18:00 winter	One-page **monthly PDF report** (account vs 60/40) posted to Discord.
First trading day, 10:45 summer / 09:45 winter	**Rebalance:** sells first, then buys, as market orders using fractional shares (whole shares for any ETF that can't be fractional). Fills and slippage logged.
First trading day, after the close	**Monthly review** against the evaluation scorecard.
Rest of the month	Nothing. No stops, no intra-month trades, no reacting to news.

Routine schedules are stored in UTC, so clock times shift by an hour when US daylight saving ends or starts. Both versions are safely after the open or the close. Holidays are handled by Alpaca's market calendar.

Who does what

Part	Role
scripts/raam.py	All the maths, every guardrail and every order. Deterministic: same data in, same answer out.
Claude (cloud routines)	**Operator, not strategist.** Runs the commands, writes the logs, does research for context, sends the Discord messages. It may halt if something looks wrong. It never changes targets, picks, timing or rules.
GitHub repository	The only persistent memory: config, rules, signals, trade log, reports. Every run starts from a fresh copy and pushes its results back.
Alpaca	Paper brokerage account and market data (a move to Interactive Brokers paper is planned after a feasibility check).
Discord	Where every message, report and alert arrives.
The owner	The only one who can change the rules, in an interactive session, with the change and its evidence written into the change log.

Principles

- **A missed rebalance is cheap; a wrong trade is not.** When data is stale, a fund changes, weights don't add up or the broker errors, the system halts and alerts instead of improvising. (Stress tests: filling up to 5 days late made no meaningful difference.)
- **Research never changes the allocation.** The one exception is a structural problem with an ETF (closure, merger, ticker change, trading halt), which is grounds to halt.
- **Keys live only in the cloud environment's variables**, never in files, commits or messages.

5 Safety rails

Checks that sit outside the strategy and can only stop trading, never add risk.

Circuit breaker

- **Trips** when the whole portfolio is **20% below its high-water mark**, or falls **5% in a single day**.
- **Action:** the next morning, everything (Core included) moves to 100% BIL.
- **Re-entry:** automatic at the next monthly rebalance, following a fresh signal, with the high-water mark reset.
- **History:** never tripped in the 2015–2026 backtest (worst daily drawdown −14.1%). In the 2008 test it tripped once (23 Feb 2009, −20.2%) and re-entered in March.
- **Trade-off:** in a one-day crash it locks the loss and can miss a fast rebound until the next month. It is a tail-risk net, not a return enhancer.

Checks before every signal and trade

Check	If it fails
Every ETF has a fresh price bar for the signal date	Halt: no signal is saved
Momentum, volatility and correlation are valid numbers	Halt
Target weights add up to 100% (±0.5%); none negative	Halt
Only universe tickers can be traded	Halt; stray positions are left alone and flagged

No single buy above 35% of the account	Halt
Account is a paper account (live needs two explicit switches)	Refuses to run
Market is open at order time	Halt; nothing submitted
Each ETF is tradable on Alpaca	Halt; non-fractionable ETFs use whole shares
Buys never exceed available cash	Buys scaled down to fit
Every order ends 'filled'	Halt and alert, with the fills recorded
The same month can't be traded twice	Second run does nothing

Tested failure drills

Automated tests simulate a rejected order, a partial fill, a broker outage, missing, stale and zero prices, an untradable ETF and an unexpected position. All end in a safe halt with no wrong trade. Two small gaps were found and are open for a decision:

- **F1:** if the broker fails halfway through a rebalance, sells that already filled are not written to the trade log (nothing wrong is traded; the record is incomplete).
- **F2:** if a day's closing price arrives blank rather than missing, that ETF is ranked on the previous day's price instead of halting (unlikely with Alpaca; small impact).

6 Historical results

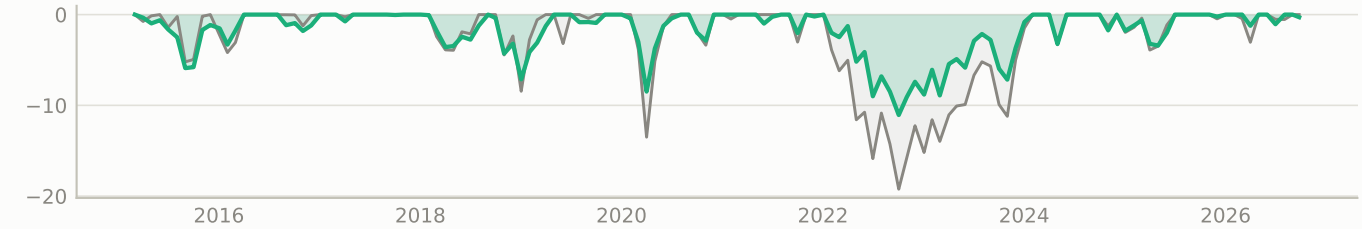
What the rules would have done from February 2015 to September 2026.

How the test works: the live script's own signal code, run at every month-end on Yahoo Finance adjusted prices; fills at the next day's open; 5 basis points of cost per trade; the circuit breaker checked daily. IBIT only exists from January 2024, so before that it is simulated from the Bitcoin price minus IBIT's 0.25% fee. Past results do not guarantee future results.

Growth of $100,000 (log scale)



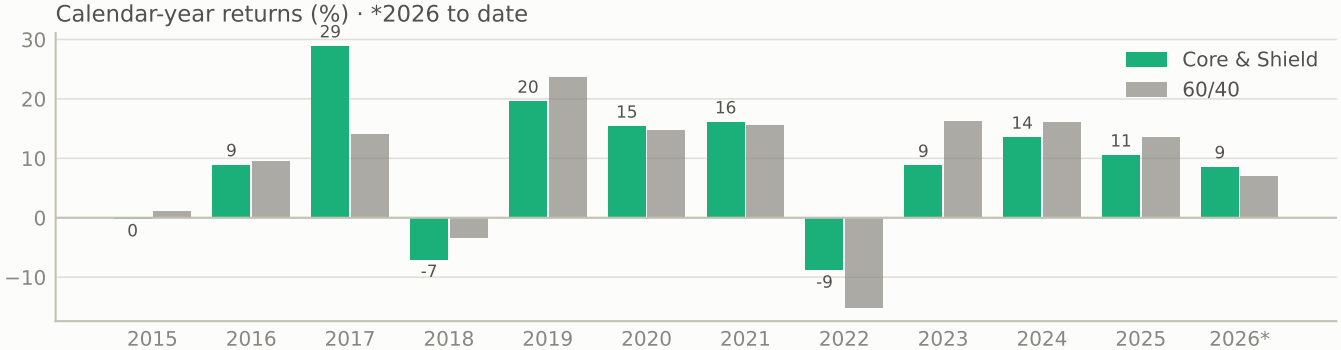
Drawdown from peak, month-end (%)



	Return / yr	Volatility	Sharpe	Worst month-end drop	Worst daily drop	2022
Core & Shield	9.3%	7.8%	1.18	−11.1%	−14.1%	−8.8%
60/40 (SPY/AGG)	9.2%	10.0%	0.93	−19.2%		−15.2%
SPY	14.1%	15.6%	0.93	−23.3%	−33.7%	−18.0%
Core & Shield without Bitcoin	6.5%	6.9%	0.95	−11.1%	−14.6%	≈ −8%

Sharpe ratio = return ÷ volatility (no risk-free rate). 2008 test (no Bitcoin, which didn't exist): Core & Shield −7.4% vs 60/40 −17.7%.

Calendar-year returns (%) · *2026 to date



Reading it: level with 60/40 over the whole period; clearly ahead in 2017 (Bitcoin) and 2022, slightly ahead in 2020, 2021 and 2026, behind in strong equity years (2019, 2023–2025). 2018 was worse than 60/40 mostly because Bitcoin's crash hit while it was held (without Bitcoin it was about level).

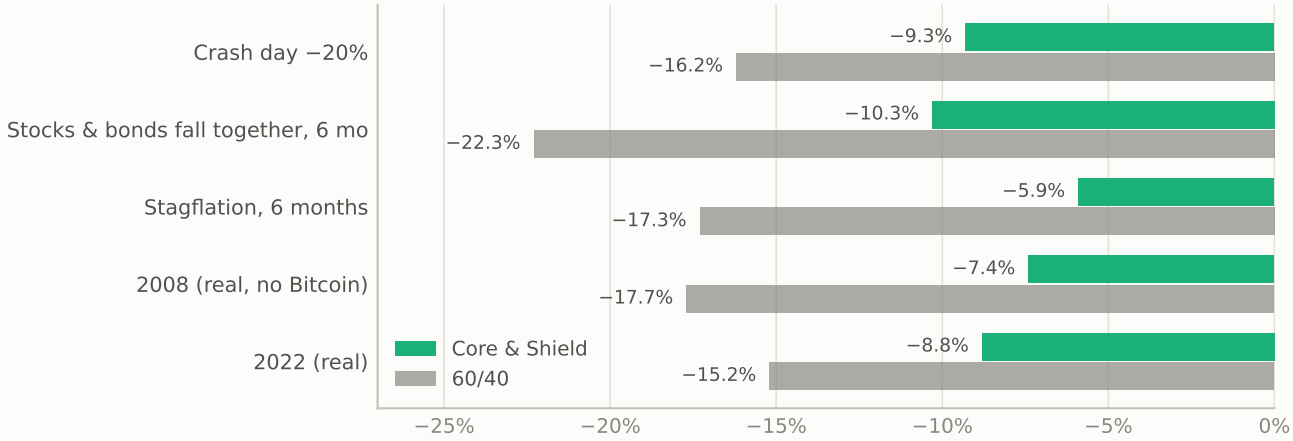
7 Stress tests

How fragile is the result? Run 27 Sep 2026 with backtest/stress.py; none of it changed the strategy.

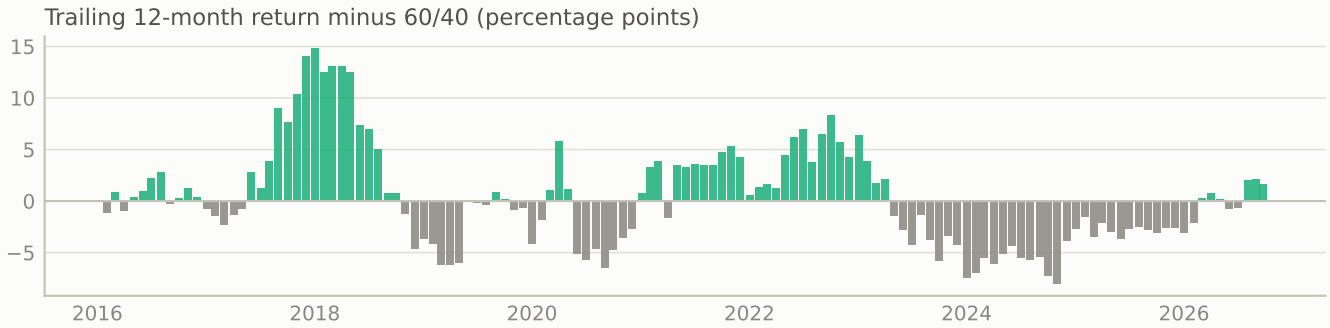
Test	What was tried	Result
Trading costs	5, 10, 25, 50 bps per trade	Risk-adjusted edge survives even 50 bps (Sharpe 0.97 vs about 0.92). The return edge disappears above about 10 bps.
Settings	13 nearby settings: lookbacks 63–126 days, hold 4 or 6, λ 0.92/0.97, each weight ±10 pp	**13 of 13** still beat 60/40's Sharpe (range 1.03–1.28). Not a lucky peak.
Late fills	Trade 2, 3 or 5 days after the signal	No real difference (Sharpe 1.18–1.23).
Any single year	127 rolling 12-month windows	Beat 60/40 on return in only **49%**, but had the **smaller drawdown in 89%**, and did better in 93% of the windows where 60/40 lost money.
Monte Carlo	5,000 simulated 10-year paths (6-month blocks of history)	**94%** chance of a higher Sharpe than 60/40, **97%** of a smaller worst drop. Any single year: 52% chance of trailing 60/40, 14% chance of a loss.

Shocks worse than anything in the test window

Each made-up shock was injected into real prices at six different dates and the whole model re-run, so it reacts as it would live. Averages shown; real 2008 and 2022 added for comparison.



- Core & Shield took **34–57% of 60/40's loss** in every made-up shock.
- **Crash day:** the −5% daily breaker trips every time, so the portfolio sits in cash until the next rebalance and can miss a quick rebound.
- **Realistic worst case:** a six-month stock-and-bond slide running straight into a Covid-style crash reached **−21%** and tripped the −20% breaker. That is deeper than anything in the backtest (−14.1%), so plan for a 20% drawdown being possible.
- **Bitcoin −80% while held:** about 3 percentage points extra loss, as the 5% cap intends.



Green = ahead of 60/40 over the previous 12 months. Long stretches behind are normal in bull markets.

8 What to expect, and how to judge it

The criteria were fixed before the first trade so results can't move the goalposts.

Strengths and weaknesses

Good at	Weak at
Slow bear markets and rising-rate years (2022: −8.8% vs −15.2%)	Strong equity bull markets: it trails SPY by a lot and 60/40 by a little
Smaller drawdowns in almost every period (89% of rolling years)	Sudden one-day crashes: a monthly model can't react in time
Robust to costs, settings and late trades	Return edge over 60/40 is thin; about half of it depends on Bitcoin
Simple, transparent, rules-based, low turnover	Single years are a coin flip against 60/40 on return

Normal ranges (from the backtest) for judging a live month

Measure	Normal	Unusual / investigate
Monthly return	Between −3% and +3% (78% of months); 65% of months positive	Outside −5.7% to +6.9%
Cash (BIL) weight	About 16% on average; about 37% in a year like 2022	Above 50%
Drawdown from peak	Up to about −11% month-end, −14% daily	Worse than −14.1%
Share of 60/40's loss in its down months	About 63%	Above 85% on average
Bitcoin (IBIT)	Held in about half of months, at 5%	Above 5% at a rebalance
Turnover	About 15% of the portfolio per month	—

Checkpoints

When	Question	Passes when
1 month (early Nov 2026)	Does it run?	All operations criteria pass for October: signal and rebalance on time, all orders filled, slippage ≤ 10 bps, holdings within ±2 pp of target, IBIT ≤ 5%, report delivered

3 months (early Jan 2027)	Is it reliable?	Operations pass 3 months out of 3; returns, cash and drawdown inside the normal ranges
6 months (early Apr 2027)	Does it behave as designed?	Live returns match a backtest of the same months within 0.3 pp a month; defence assessed if 60/40 had 2+ down months
12 months (early Oct 2027)	Is it worth running?	See the verdict rules below

The 12-month verdict

- **Working as designed:** operations pass in 11+ months, behaviour matches the backtest, and the 12-month drawdown is smaller than 60/40's (plus the down-month test, if 60/40 had 2+ down months).
- **Inconclusive:** everything passes but 60/40 had fewer than 2 down months. Normal in a calm year; the defence hasn't been tested yet. Keep running.
- **Not working:** the live account doesn't match the model, or it takes more than 100% of 60/40's losses, or its drawdown is larger than 60/40's in a year when 60/40 fell.
- **Explicitly not a failure:** trailing SPY, or trailing 60/40 in a year when 60/40 rose.

> **Rule for the evaluation year:** don't change the strategy unless something is broken. A change resets the evidence. Ideas are parked and tested as fixed variants at the 6-month checkpoint.

9 Decision history

How the strategy got here, including what was tried and rejected.

Date	Decision	Evidence
23 Sep 2026	Set up pure RAAM from the paper, with defaults for its 7 ambiguities	Paper's +638% claim could not be reproduced
23 Sep 2026	**Tier 1 upgrades:** cash test uses BIL's momentum (was > 0%); BIL replaces SHY as cash; breaker at −20% with automatic re-entry (was −15%, manual)	Safer behaviour at little cost
23 Sep 2026	**Trend factor removed** from the score	Three backtests agreed: better return, smaller drawdown, in both halves of the period
24 Sep 2026	**Core & Shield:** 50% fixed 60/40 + 50% RAAM	RAAM alone lagged badly in bull markets; the blend matched 60/40's Sharpe with far smaller drawdowns
24 Sep 2026	Strategy locked for a 12-month paper evaluation	Owner decision
Sep 2026	Trade the morning after the signal (10:45 ET summer / 09:45 winter)	Month-end close, first day and mid-month compared; no major downside to keeping it
26 Sep 2026	**IBIT added**, capped at 5% of the portfolio	Return 6.5% → 9.3%, Sharpe 0.95 → 1.18, worst drop slightly better; better in every sub-period; about half the gain survives with Bitcoin's long-run return flattened to 0%
26 Sep 2026	Evaluation criteria written and fixed	Before any live result existed

27 Sep 2026	Stress tests run (section 7)	Robust to costs, settings and timing; worst case about −21%

Tested and rejected, or parked

Idea	Verdict	Why
Swap VV→TLT and VAW→GLD (crisis assets)	Rejected	Better before 2020, worse after; made March 2020 and 2008 worse
Correlation measured against the Core too	Rejected	Worse from 2020 onward
Size positions by inverse volatility	Rejected	Failed from 2020 onward
Heavier momentum weight	Not recommended	Trades defence for return
Bitcoin uncapped (10% slots)	Not adopted	12.7%/yr but more dependent on Bitcoin
Volatility-managed Core	Parked	Shallower losses in every crisis, same Sharpe, about 0.7%/yr less return
Risk-based sizing, staggered rebalancing	Parked	Candidates for the 6-month checkpoint

Honest assessment at launch

Rated about **70/100** before the first trade: a sound, defensive, well-guarded balanced strategy with a thin return edge over 60/40, a real but modest dependence on Bitcoin, and no live track record yet. About 11 years of data means a Sharpe estimate is only accurate to roughly ±0.3, so most differences between variants are not statistically significant. The live year is what decides it.

Glossary

Term	Meaning
60/40	60% US stocks (SPY) + 40% US bonds (AGG): the benchmark
Sharpe ratio	Return per unit of volatility; higher is better. 1.0+ is good for a balanced fund
Drawdown	Fall from the highest value reached so far
Momentum	How much a price has changed over a lookback period (here about 4 months)
Basis point (bp)	0.01%. 10 bps = 0.1%
Slippage	Difference between the price when an order was planned and the price it filled at
Circuit breaker	Automatic move to cash after a −20% drawdown or −5% day
High-water mark	The highest account value so far, used to measure drawdown